A first order universal growth chart on the $s = 0 + I \cdot t$ line for periodic 1st degree Dirichlet character L-functions.

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Executive Summary

A comparison of the extreme peak height behaviour of several periodic 1st degree L-functions along with some Davenport-Heilbronn functions is presented on the $s = 0 + I \cdot t$ line (and for the Ramanujan Tau L-function on the $s = 5.5 + I \cdot t$ line) i.e., the lower boundary of the L-function critical strip. It is empirically demonstrated (similar to observed critical line behaviour) that a first order universal growth chart on the critical line for periodic 1st degree L-functions can be constructed by (i) rescaling of the t co-ordinate and (ii) rescaling of the L-function height by the proportion of zero (periodic) dirichlet characters. In addition, a Box Cox transformation model is fitted to the known extreme peak heights $|Z|$ of the Riemann Zeta function on the $s = 0 + I \cdot t$ line to yield a extreme peak trend growth estimate for periodic L-functions with $|\chi| = 0, 1$ dirichlet characters on the lower boundary of the critical strip. The trend model performance is then assessed by mapping the model from $s = 0 + I \cdot t$ to $s = 1 + I \cdot t$ (using the Riemann Zeta functional equation) which helps highlight deficiencies in the trend model.

Introduction

Various authors [1-21] have identified large and/or extreme Riemann Siegel Z function peaks. Most recently, Martin [21] has reported extreme Riemann Siegel Z function peaks for a range of 1st degree L functions and some of their Davenport-Heilbronn counterparts on the critical line $s = 0.5 + I \cdot t$ (as well as the Ramanujan Tau L-function on its critical line $s = 6 + I \cdot t$) using for progressively higher t values (i) available high precision L-function calculations, (ii) high precision tapered L-function dirichlet series estimates at the second quiescent region $N_2 = 2 \cdot \left(\frac{t}{2\pi}\right)^d \cdot N_C$, (iii) multiple decimal place accuracy tapered first order Riemann Siegel formula L-function estimates at the first quiescent region $N_1 = \sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C}$, (iv) multiple decimal place accuracy spectrally filtered Euler Product L-function analogues of the first order Riemann Siegel formula as well as (v) 10% accuracy (critical line), 1% accuracy (s=0,1 lines) preliminary peak height estimates for t < 1e30 using a spectrally filtered heavily truncated Euler Product method $N_{sub1} = \beta \cdot 0.5 \cdot \left(\sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C}\right)^{0.5}$ where $\beta = N_C$ for 1st degree L-functions. Where overlap of adjoining methods was included to validate calculation performance.

In particular, Martin [21] noted that (i) universal growth chart behaviour of extreme peak behaviour of 1st degree L-functions with {0,1} magnitude dirichlet characters (in first order agreement with the Riemann Zeta Riemann Siegel |Z| function behaviour) can be observed by changing the x-axis (of such 1st degree L-functions) from $t \to \log\left[\sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C}\right]$ and adjusting the y axis $log(|Z|) \to log(|Z| \cdot \frac{N_C}{N_{\chi \neq 0}})$ and (ii) an empirical trend growth estimate can be obtained for known Riemann Zeta Riemann Siegel |Z| function 1st degree L-functions with {0,1} magnitude dirichlet characters extreme peaks on the critical line for t < 1e40 via the fitted Box Cox transformation model $log\left[|Z_L(k+I\cdot t)|_{max}\cdot\frac{N_C}{N_{\chi\neq 0}}\right]_{trend} \lesssim 1.274014+0.866363\cdot\left(\frac{\sqrt{\log\left[\sqrt{\left(\frac{t}{2\pi}\right)^d\cdot N_C}\right]}-1}{0.5}\right).$

In this paper, (i) the extreme peak height growth behaviour along the $s=0+I\cdot t$ line of the Riemann Siegel Z function analogues of

- $\zeta(0.0+I\cdot t)$ [22-26],
- $L(\chi_3(2,.),0.0+I\cdot t)$ [27],
- $L(\chi_4(3,.),0.0+I\cdot t)$ [27],
- $L(\chi_5(4,.),0.0+I\cdot t)$ [27],
- the dual pair $L(\chi_5(3,.),0.0+I\cdot t)$ [27], $L(\chi_5(2,.),0.0+I\cdot t)$ [27] and

  their Davenport-Heilbronn function counterparts $f_1(0.0+I\cdot t)$ and $f_2(0.0+I\cdot t)$ [28-30],
- $L(\chi_7(6,.),0.0+I\cdot t)$ [27],
- the dual pair $L(\chi_7(5,.),0.0+I\cdot t)$ [27], $L(\chi_7(3,.),0.0+I\cdot t)$ [27], and

  their Davenport-Heilbronn function counterparts $f_3(0.0+I\cdot t)$ and $f_4(0.0+I\cdot t)$ [31],
- $L(\chi_37(36,.),0.0+I\cdot t)$ [27],
- $L(\chi_51(50,.),0.0+I\cdot t)$ [27]

and $s=5.5+I\cdot t$ line of the Riemann Siegel Z function analogue of

- the Ramanujan Tau L-function 2-1-1.1-c11-0-0 which has degree 2 and character $\chi_1(1,.)$ [27,32].

are compared on the lower boundary of their critical strip to further understand the growth behaviour of L-functions and the Riemann Zeta Rieman Siegel Z function, (ii) Box Cox transformation modelling (and simpler log(|Z|) vs log(t) modelling) of the Riemann Zeta function extreme peak Riemann Siegel Z function growth is conducted to provide a trend growth prediction on the critical strip lower boundary (i.e. extreme peak heights occur either side of the trend growth line) and (iii) the fitted $s=0+I\cdot t$ trend growth estimate is used with the Riemann Zeta functional equation to estimate the $s=1+I\cdot t$ (critical strip upper boundary) extreme peak trend behaviour for t < 1e40.

In contrast to [21], where for the 2nd degree Ramanujan Tau L-function just methods (i-iii) were employed as the interval investigated was only t < 6600, in this paper, extreme peaks for the wider interval t < 1e13 were identified for the 2nd degree Ramanujan Tau L-function and it has been empirically observed when using the spectrally filtered heavily truncated Euler Product method that $\beta_{tauL}=100,10,1$ hyperparameter settings for the 2nd degree Ramanujan Tau L-function extreme peak estimates were achieving ~1%, ~10%, weaker accuracy respectively on its critical line using method (iii) for benchmarking.

Combined comparison of results for 1st degree L-functions

In figures 1 and 2, the extreme peak growth behaviour of the above 1st degree L-functions (all with dirichlet character magnitudes $|\chi|=0,1$) are compared in the range 0 < t < 1e20.

- When plotting log(|Z|) vs log(t) in figure 1 it can be seen that each L-function varies in their growth rates of the extreme peak heights as a function of log(t). They are similar but different.

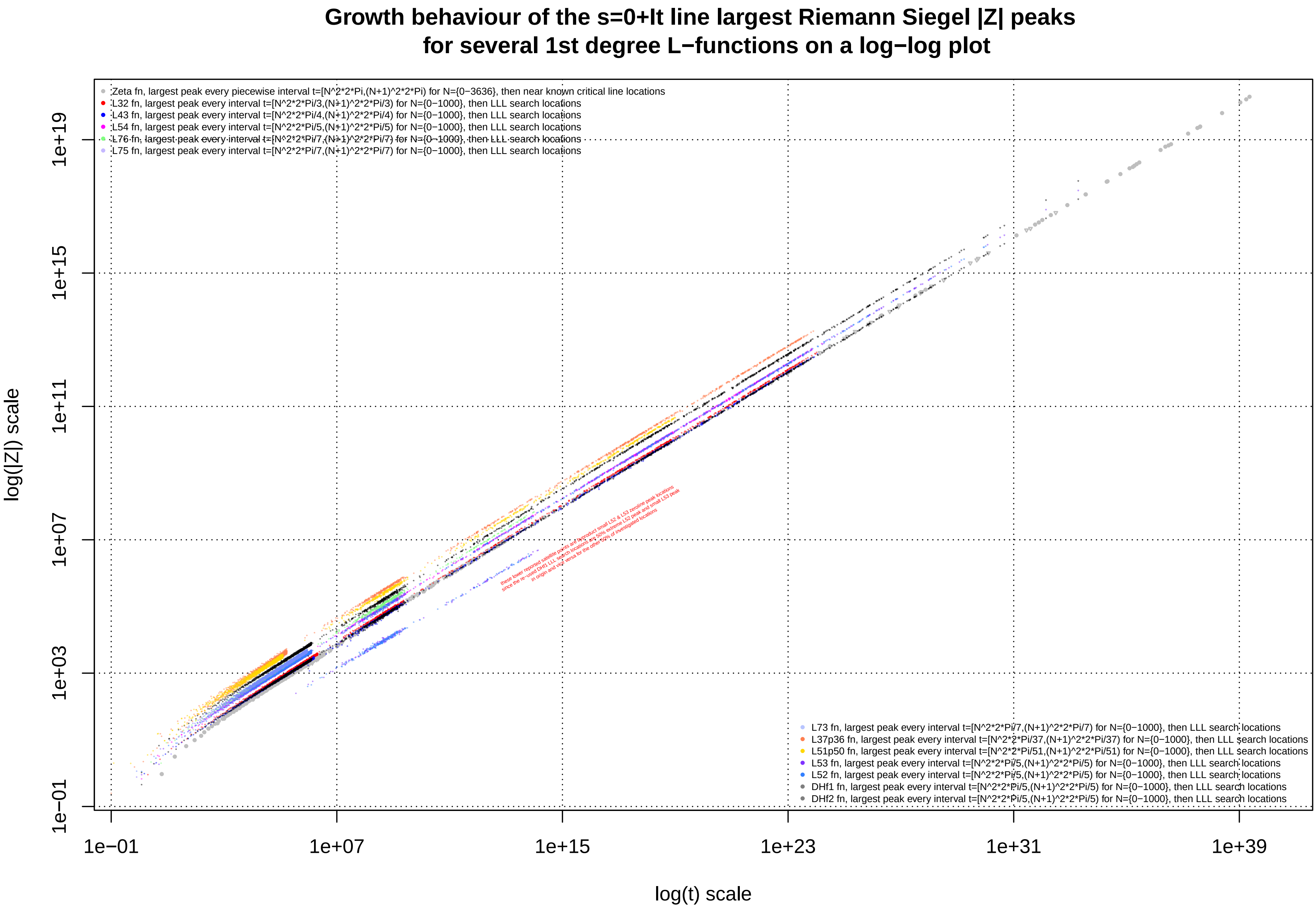

Figure 1: Standard log-log overlay of the growth behaviour of extreme Riemann Siegel Z function peak heights |Z| for different 1st degree L-functions in the interval t < 1e31.

- However in figure 2 by

  1. changing the x-axis from $log(t) \to \log\left[\sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C}\right]$ AND

  2a. adjusting the y axis $log(|Z|) \to log(|Z| \cdot \frac{N_C}{N_{\chi \neq 0}})$ for L-functions OR

  2b. adjusting the y axis $log(|Z|) \to log\left(|Z| \cdot \left\{\frac{N_C}{\sum_{k=1}^{N_C} |\chi_k|}\right\}\right)$ for Nc-periodic Davenport-Heilbronn functions

  where N_C is the L-function conductor, $N_{\chi \neq 0}$ is the number of non-zero dirichlet characters Mod N_C and $|\chi_k|$ is the magnitude of the periodic dirichlet characters, it can be seen that under these rescaled xy axes the 1st degree L-functions growth rates coincide reasonably well with the (absolute value |Z|) Riemann Zeta Riemann Siegel Z function behaviour as $t \to \infty$.

Such behaviour (transformation of different L-functions to a universal reference frame) relies on the strong piecewise interval behaviour $\left(N_1 = \sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C}\right)$ of L-functions with {0,1} magnitude dirichlet characters depending on the L-function conductor value N_C & degree d in the Riemann Siegel formula and the bounded (Dirac comb) nature occurring at extreme 1st-degree L-function peaks (i.e. $|\Im(\frac{\chi}{n^s})| \lesssim \pm|\chi| \lesssim \pm 1$ for $\chi \neq 0$ and $\frac{\chi}{n^s} = 0$ for $\chi = 0$ for the low lying Dirichlet series terms $\frac{\chi_L}{n^s}$).

That is, for the subset of L-functions with dirichlet characters $|\chi| = 0, 1$ therefore each $|\Im(\frac{\chi}{n^s})| \lesssim \pm|\chi| \lesssim \pm 1$ for $\chi \neq 0$ and so each dirichlet term only contribute a maximum of 1 to |Z| summing up to the dirichlet series truncation point $N_1 = \sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C}$ (or $N_2 = 2 \cdot \left(\frac{t}{2\pi}\right)^d \cdot N_C$) depending on which approximation approach i.e., methods (iii), (v) (or (ii), (iii)) mentioned in the introduction are used. Hence the extreme Riemann Siegel function peak value for the subset of L-functions with dirichlet characters $|\chi| = 0, 1$ will differ in first order from the Riemann Zeta function behaviour by the proportion of dirichlet characters that are non-zero when compared on a common x-axis scale i.e. $log(\sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C})$.

By close examination of the ordered source data or by magnifying figures 1 and 2 it can be seen that the $s = 0 + I \cdot t$ Rieman Siegel |Z| function extreme peak heights are not strictly monotonically increasing with t but the trend growth component is clearly dominant. Model fitting results of the trend growth rate of the extreme peak height will be presented later in this paper.

Also present in figures 1 & 2 are some low height peaks (i.e., not extreme) for $L(\chi_5(3,.), 0.0 + I \cdot t)$ & $L(\chi_5(2,.), 0.0 + I \cdot t)$ and ($L(\chi_7(5,.), 0.0 + I \cdot t)$ & $L(\chi_7(3,.), 0.0 + I \cdot t)$) L-functions because the search method for predicting extreme peaks for these dual pair L-functions relied on pre-screen calculations of the behaviour of the Davenport-Heilbronn functions $f_1(0.0 + I \cdot t)$ and ($f_4(0.0 + I \cdot t)$) respectively. That is, half each of the $f_1(0.0 + I \cdot t)$ extreme peaks are due to $L(\chi_5(3,.), 0.0 + I \cdot t)$ ($L(\chi_5(2,.), 0.0 + I \cdot t)$) extreme peaks and likewise half each of the $f_4(0.0 + I \cdot t)$ extreme peaks are due to $L(\chi_7(5,.), 0.0 + I \cdot t)$ ($L(\chi_7(3,.), 0.0 + I \cdot t)$) extreme peaks. So when $L(\chi_5(3,.), 0.0 + I \cdot t)$ ($L(\chi_5(2,.), 0.0 + I \cdot t)$) grid search or euler product spectrum calculations are run over the $f_1(0.0 + I \cdot t)$ identified extreme peak locations only half of the $L(\chi_5(3,.), 0.0 + I \cdot t)$ ($L(\chi_5(2,.), 0.0 + I \cdot t)$) peaks are extreme and likewise when $L(\chi_7(5,.), 0.0 + I \cdot t)$ ($L(\chi_7(3,.), 0.0 + I \cdot t)$) grid search or euler product spectrum calculations are run over the $f_4(0.0 + I \cdot t)$ extreme peaks locations.

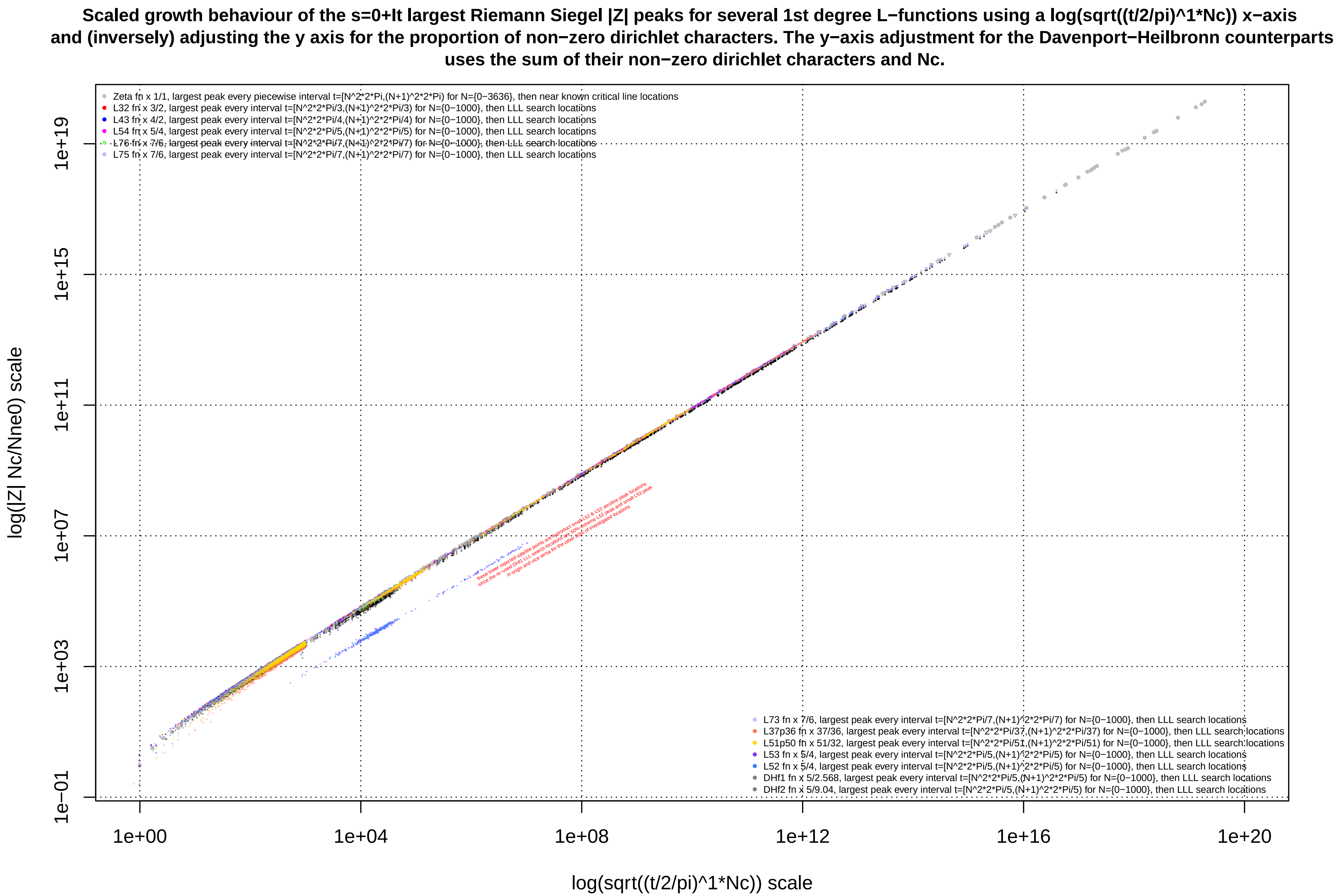

Figure 2: Rescaled XY axes overlay of the growth behaviour of extreme Riemann Siegel Z function peak heights |Z| for different 1st degree L-functions in the interval t < 1e20. The |Z| values on the Y axis are scaled by the ratio of the (L-function conductor value)/(number of non-zero dirichlet characters mod Nc) i.e. $log(|Z| \cdot \frac{N_C}{N_{\chi \neq 0}})$. (For the 5-periodic Davenport-Heilbronn functions $log(|Z| \cdot \frac{5}{\sum_{k=1}^{5} |\chi_k|})$ is used for Y axis rescaling.) The t co-ordinate on the X axis is transformed to the first quiescent region value of the L-function i.e. $log(\sqrt{(\frac{t}{2\pi})^1 \cdot N_C})$.

Individual comparison of results for 1st degree L-functions

Searching and identifying candidate locations of extreme Riemann Siegel |Z| function peaks for L-functions

In similar manner to [19-21], for each given L-function (and Davenport-Heilbronn counterpart) extreme Riemann Siegel |Z| function peaks in the range 0 < t < 1e30 on the critical line using for progressively higher t values (i) available high precision L-function calculations, (ii) high precision tapered Riemann Zeta dirichlet series estimates at the second quiescent region $N_2 = 2 * \left(\frac{t}{2\pi}\right)^d * N_C$ (iii) multiple decimal place accuracy tapered first order Riemann Siegel formula estimates at the first quiescent region $N_1 = \sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C}$, (iv) multiple decimal place accuracy spectrally filtered Euler Product analogues of first order Riemann Siegel formula as well as (vi) preliminary peak height estimates using a spectrally filtered heavily truncated Euler Product method $N_{sub1} = \frac{N_C}{2} \cdot \left(\sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C}\right)^{0.5}$. Again with overlap of adjoining methods was included to validate calculation performance. Following [21], $\beta = N_C$ rather than $\beta = 1$ was used for the heavily truncated N_{sub1} method.

1. To identify extreme Riemann Siegel |Z| function peak locations for each given L-function,

1. a grid search $\Delta = 0.05$ for $t < 26^2 \cdot (2 \cdot \pi)^d / N_C$ within each $t = [N^2 \cdot (2 \cdot \pi)^d / N_C, (N+1)^2 \cdot (2 \cdot \pi)^d / N_C)$ interval using estimation methods (i) is conducted with some manual cleanup for peaks on the boundary edges between piecewise intervals
2. a grid search $\Delta = 0.05$ for $26^2 \cdot (2 \cdot \pi)^d / N_C \leq t < 65^2 \cdot (2 \cdot \pi)^d / N_C$ within each $t = [N^2 \cdot (2 \cdot \pi)^d / N_C, (N+1)^2 \cdot (2 \cdot \pi)^d / N_C)$ interval using estimation method (ii) with a method (i) calculation for the final identified peak location
3. a grid search $\Delta = 0.1$ for $65^2 \cdot (2 \cdot \pi)^d / N_C \leq t < 1001^2 \cdot (2 \cdot \pi)^d / N_C$ within each $t = [N^2 \cdot (2 \cdot \pi)^d / N_C, (N+1)^2 \cdot (2 \cdot \pi)^d / N_C)$ interval using estimation method (iii) with a method (ii) calculation for the final identified peak location
4. using LLL algorithm predictions for extreme peak locations

    (I) a short interval grid search $\Delta = 0.05$ over a $\delta t = \pm 0.25$ interval about the LLL predicted location using estimation method (iii) is conducted (with some use of method (ii) for lower t values) and/or

    (II) a short 256 point fourier grid search $\Delta = 0.05$ over a $\delta t = [-6.35, 6.4]$ interval about the LLL predicted location using estimation method (iv) (spectral filtered Euler Product N_1 truncation) often conducted with a method (iii) calculation for the final identified peak location

    (III) for $t \gtrsim 10^{15}$ a short 256 point fourier grid search $\Delta = 0.05$ over a $\delta t = [-6.35, 6.4]$ interval about the LLL predicted location using preliminary estimation method (v) (spectral filtered Euler Product with heavy N_{1sub} truncation)

2. Adjusting the LLL search algorithm [33] constraints for periodic 1st degree L-functions with non-zero dirichlet characters with magnitudes of order unity ($|\chi| = 1$)

The LLL search algorithm for each investigated periodic 1st degree L-function employed the $\frac{t_{peak} * log(k \bmod N_C = \{0, \ldots, N_C - 1\})}{2\pi}$ constraints identified by [21] which appears to be related to the denominator of the fractional exponents of the $(N_C - 1)$th cyclotomic polynomial roots of unity plus the N_Cth constraint $\frac{t_{peak} * log(k \bmod N_C = 0)}{2\pi} = 0$ (where in practice, the last constraint drops particular integers from the LLL algorithm calculation).

The above prescription does successfully identify candidate locations of extreme peak heights. However, that does not mean the above prescription definitively describe all constraints for possible extreme peaks nor will an actual implemented LLL algorithm report all possible instances of extreme peak locations.

Box Cox transformation model of linear regression fit of trend growth in extreme |Z| peak height for 1st degree periodic L-functions along $s = 0 + I \cdot t$

Box Cox transformation modelling [37,38] is conducted to approximate the non-linearity in the growth curve in figure 2. Under the suspicion that the non-linearity in figure 2 is approximately of the form $\log\left(\sqrt{(\frac{t}{2\pi})^d \cdot N_C}\right)$ the input linear regression model and the Box Cox transformation model output diagnostics in figure 3 both given below provides a reasonable fit of the trend growth behaviour to $\frac{(x^{0.9604378}-1)}{0.9604378}$ Box Cox transformation behaviour. For example, the intercept and linear regression coefficients are highly statisticallly significant (and given in equation (1) below) and the mode of the Box Cox transformation fit in figure 3 is $\lambda = 1$ which is evidence for a linear polynomial fit (with respect to $\frac{(\left(\sqrt{\log(\sqrt{(\frac{t}{2\pi})^1 \cdot 1})}\right)^{0.9604378}-1)}{0.9604378}$) for the Riemann Zeta function extreme peaks shown in figure 2.

Yielding the following phenomenological model of trend growth in extreme $|Z_\zeta(0 + I \cdot t)|$ peak height on the critical line based on high quality data for t < 1e31,

$$\log\left[|Z_\zeta(0 + I \cdot t)|_{max}\right]_{trend} \approx 2.1656076 + 1.1459223 \cdot \left(\frac{\left(\log\left[\sqrt{\frac{t}{2\pi}}\right]\right)^{0.9604378} - 1}{0.9604378}\right) \tag{1}$$

which can be potentially generalised based on figure 2, for **1st degree periodic L-functions with non-zero dirichlet characters with magnitudes of order unity ($|\chi| = 1$)** to

$$\log\left[|Z_{L,d=1}(0+I\cdot t)|_{max}\cdot\frac{N_C}{N_{\chi\neq 0}}\right]_{trend} \lesssim 2.1656076+1.1459223\cdot\left(\frac{\left(\log\left[\sqrt{\left(\frac{t}{2\pi}\right)^1 \cdot N_C}\right]\right)^{0.9604378} - 1}{0.9604378}\right) \tag{2}$$

Extrapolation of the model into the region 1e31 < t < 1e40 and beyond, however compared to the critical line case [21], really requires higher quality data and/or analytic results for t > 1e31 to properly test the veracity of the above trend growth model at higher t.

For instance a heuristic trend growth model that also performs qualitatively well in the region t < 1e40 is the simpler parametrisation

$$\log\left[|Z_\zeta(0 + I \cdot t)|_{max}\right]_{\text{trend}} \approx 2.1656 + \log\left(\sqrt{\frac{t}{2\pi}}\right) \tag{3}$$

as shown when comparing equations (1) (brown dot-dashed line) and (3) (green dashed line) in figure 4.

Of great interest with respect to the Box Cox model estimate result is that the confidence interval of magnitude of the intercept $2.1656076 \pm 1.96 * 0.0054552$ includes the known value of the exponential generating function for the harmonic numbers [39,40]

$$\sum_{n=1}^{\infty} H_n \frac{1^n}{n!} = e^1 \cdot (\gamma + \ln 1 + \Gamma(0,1)) \\ = 2.1653822153... \tag{4}$$

suggesting a measurable Euler-Mascheroni constant contribution to the trend growth rate of extreme |Z| peaks on the Riemann Zeta function $s = 0 + I \cdot t$ line.

However, as presented later in figure 13 when vetting equation (1) or (3) based predictions co-opting the Riemann Zeta functional equation for the estimated trend growth rate against observed $s = 1 + I \cdot t$ extreme peak growth behaviour model equation (3) predictions are unphysical for $s = 1 + I \cdot t$ and model equation (1) is clearly insufficient for extrapolation beyond the fitted region t < 1e31.

Linear regression model fit for $\log\left[|Z_\zeta(0 + I \cdot t)|_{max}\right]_{trend} \sim \left(\frac{\left(\log\left[\sqrt{\frac{t}{2\pi}}\right]\right)^{0.9604378} - 1}{0.9604378}\right)$

```
## [1] 1
```

```
##
## Call:
## lm(formula = yt3 ~ xt3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47472 -0.03240  0.01243  0.03783  0.15306
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2.1656076  0.0054552     397   <2e-16 ***
## xt3         1.1459223  0.0003616    3169   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06545 on 316 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 1.004e+07 on 1 and 316 DF,  p-value: < 2.2e-16
```

Summary of languages/IDEs used for calculations

The LLL search based algorithm, the high precision L-function calculations, tapered dirichlet series calculations, input discrete fourier spectra samples, the fourier transforms, the spectral filtering, the rescaled conjugate mirror reflection values and the inverse fourier transforms are calculated using Pari-GP [34] and the Box Cox transformation modelling and general graphing is performed using R [35] and RStudio IDE [36].

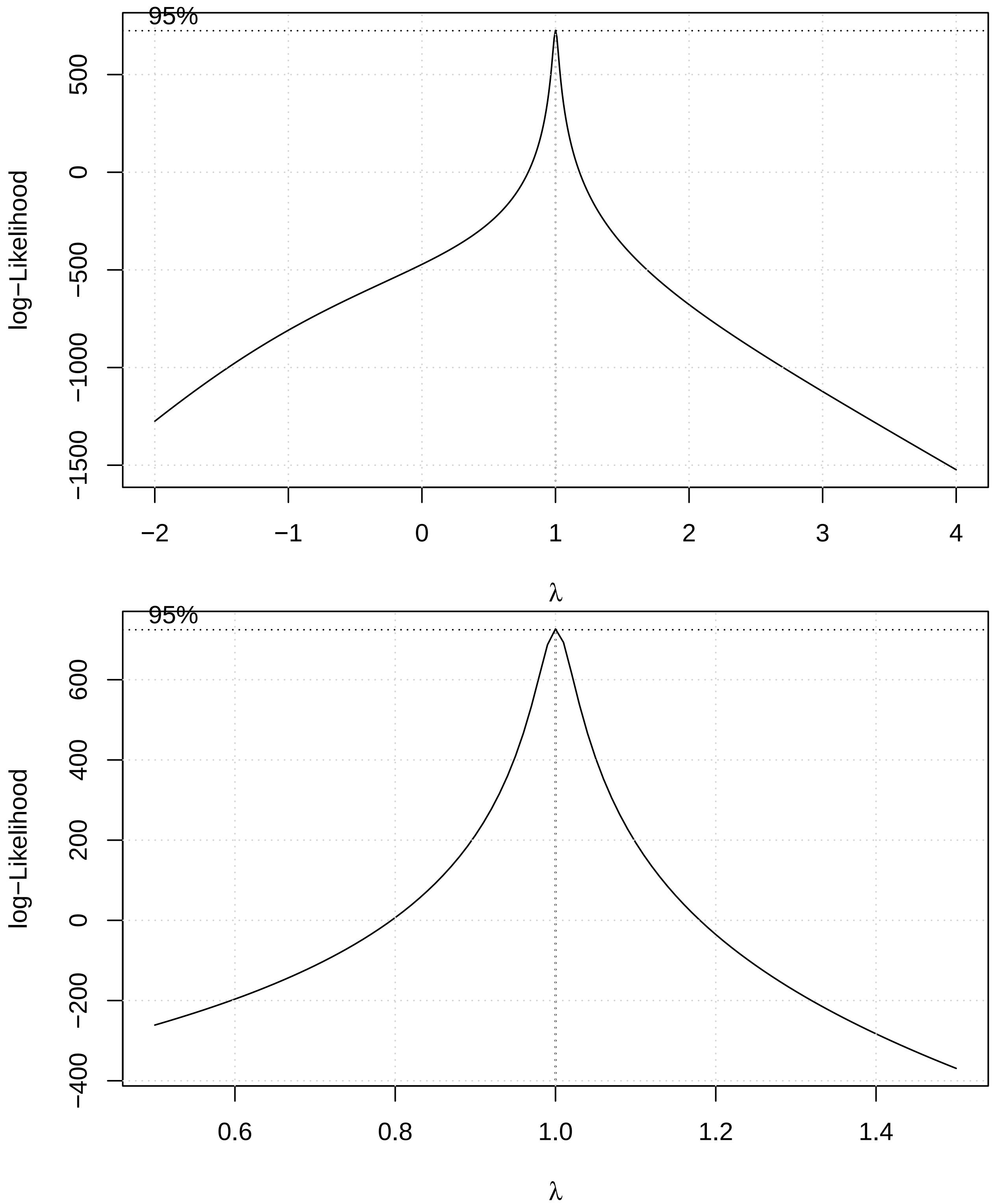

Figure 3: Box Cox modelling analysis using the MASS package on the growth behaviour of Riemann Zeta function extreme Riemann Siegel Z function peak heights |Z| for in the interval t < 1e31. Modelling Y $\sim$ X, the input data comprises Y = $log(|Z|)$ and X = $\frac{\left(log(\sqrt{(\frac{t}{2\pi})^{d} \cdot N_C})^{0.9604378}-1\right)}{0.9604378}$. Top panel - wide view of the log-likelihood of the Box Cox model fit supporting $\lambda = 1$ (i.e. linear behaviour with respect to the input regression model), Bottom panel - magnified view about $\lambda = 1$

Individual L-function and Davenport-Heilbronn function results in comparison to $|Z_\zeta(0.5 + I \cdot t)|$

Figures 4-11 below, display for each investigated 1st degree 1 L-function or Davenport-Heilbronn function

1. left column - $log\left(|Z_L|\right)$ vs $\log(t)$ of extreme L-function (or Davenport-Heilbronn function) peak height
2. right column - $log\left(|Z_L| \cdot \frac{N_C}{N_{\chi \neq 0}}\right)$ vs $\log\left(\sqrt{(\frac{t}{2\pi})^{d=1} \cdot N_C}\right)$ of extreme L-function (or Davenport-Heilbronn function) peak height

Figure 4 displays a comparison of $|Z_\zeta|$ behaviour with located extreme Riemann Siegel Z function peaks for

- top row $L(\chi_1(1,.), 0.5 + I \cdot t)$ i.e. $\zeta(0.5 + I \cdot t)$ with $N_C = 1$ and $N_{\chi \neq 0} = 1$,
- middle row $L(\chi_3(2,.), 0.5 + I \cdot t)$ with $N_C = 3$ and $N_{\chi \neq 0} = 2$,

Figure 5 displays a comparison of $|Z_\zeta|$ behaviour with located extreme Riemann Siegel Z function peaks for

- top row $L(\chi_4(3,.), 0.5 + I \cdot t)$ with $N_C = 4$ and $N_{\chi \neq 0} = 2$
- bottom row $L(\chi_5(4,.), 0.5 + I \cdot t)$ with $N_C = 5$ and $N_{\chi \neq 0} = 4$,

Figure 6 displays a comparison of $|Z_\zeta|$ behaviour with located extreme Riemann Siegel Z function peaks for

- top row $L(\chi_5(3,.), 0.5 + I \cdot t)$ with $N_C = 5$ and $N_{\chi \neq 0} = 4$,
- bottom row $L(\chi_5(2,.), 0.5 + I \cdot t)$ with $N_C = 5$ and $N_{\chi \neq 0} = 4$

Figure 7 displays a comparison of $|Z_\zeta|$ behaviour with located extreme Riemann Siegel Z function peaks for

- top row $L(\chi_7(6,.), 0.5 + I \cdot t)$ with $N_C = 7$ and $N_{\chi \neq 0} = 6$,
- middle row $L(\chi_3 7(36,.), 0.5 + I \cdot t)$ with $N_C = 37$ and $N_{\chi \neq 0} = 36$,

Figure 8 displays a comparison of $|Z_\zeta|$ behaviour with located extreme Riemann Siegel Z function peaks for

- $L(\chi_5 1(50,.), 0.5 + I \cdot t)$ with $N_C = 51$ and $N_{\chi \neq 0} = 32$

Figure 9 displays a comparison of $|Z_\zeta|$ behaviour with located extreme Riemann Siegel Z function peaks for

- top row $f_2(0.5 + I \cdot t)$ a 5-periodic Davenport-Heilbronn function with $N_C = 5$ and $\sum_{k=1}^{5} |\chi_k| \approx 9.04$,
- middle row $f_1(0.5 + I \cdot t)$ a 5-periodic Davenport-Heilbronn function with $N_C = 5$ and $\sum_{k=1}^{5} |\chi_k| \approx 2.568$

Figure 10 displays a comparison of $|Z_\zeta|$ behaviour with located extreme Riemann Siegel Z function peaks for

- top row $L(\chi_7(5,.), 0.5 + I \cdot t)$ with $N_C = 7$ and $N_{\chi \neq 0} = 6$,
- bottom row $L(\chi_7(3,.), 0.5 + I \cdot t)$ with $N_C = 7$ and $N_{\chi \neq 0} = 6$

Figure 11 displays a comparison of $|Z_\zeta|$ behaviour with located extreme Riemann Siegel Z function peaks for

- top row $f_3(0.5 + I \cdot t)$ a 7-periodic Davenport-Heilbronn function with $N_C = 7$ and $\sum_{k=1}^{7} |\chi_k| \approx 7.2077509$,
- bottom row $f_4(0.5 + I \cdot t)$ a 7-periodic Davenport-Heilbronn function with $N_C = 7$ and $\sum_{k=1}^{7} |\chi_k| \approx 4.3042576$,

As mentioned earlier also displayed in figure 3 are two models of the trend growth rate given by equations (1) and (3) shown as a brown dot-dashed and green dashed lines respectively.

For equations 4-11, only the Box Cox transformation based trend growth rate model equation (1) is displayed in the right column panels.

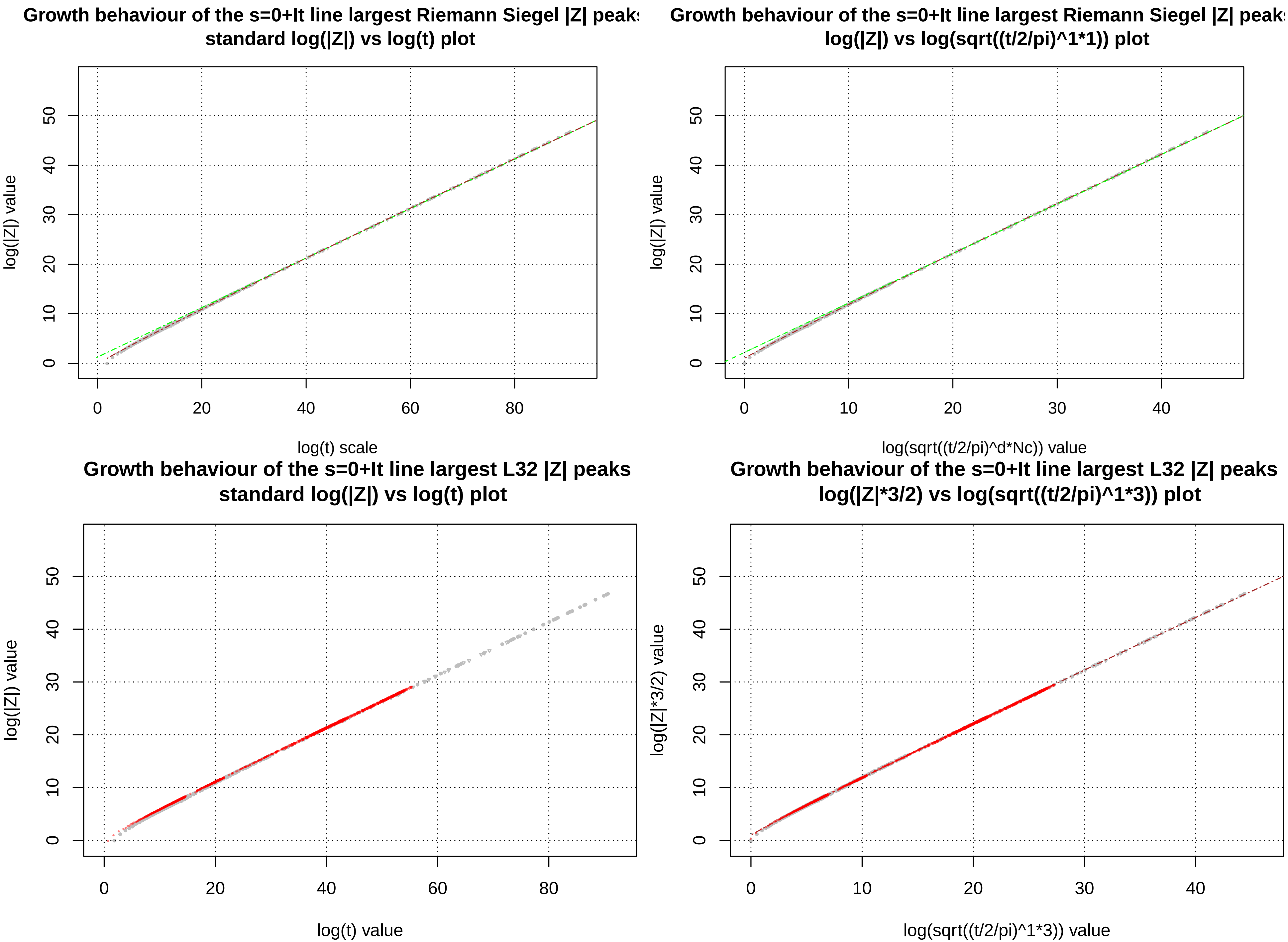

Figure 4: A graphical comparison for each investigated 1st degree L-function or Davenport-Heilbronn function of **left column** - $\log(|Z_L|)$ vs $\log(t)$, **right column** - $\log\left(|Z_L| \cdot \frac{N_C}{N_{\chi\neq 0}}\right)$ vs $\log\left(\sqrt{(\frac{t}{2\pi})^d \cdot N_C}\right)$ for top row - $L(\chi_1(1,.), 0.0 + I \cdot t)$ and bottom row - $L(\chi_3(2,.), 0.0 + I \cdot t)$. With two (one) fitted models for extreme peak height trend growth rate shown in the top (bottom) row.

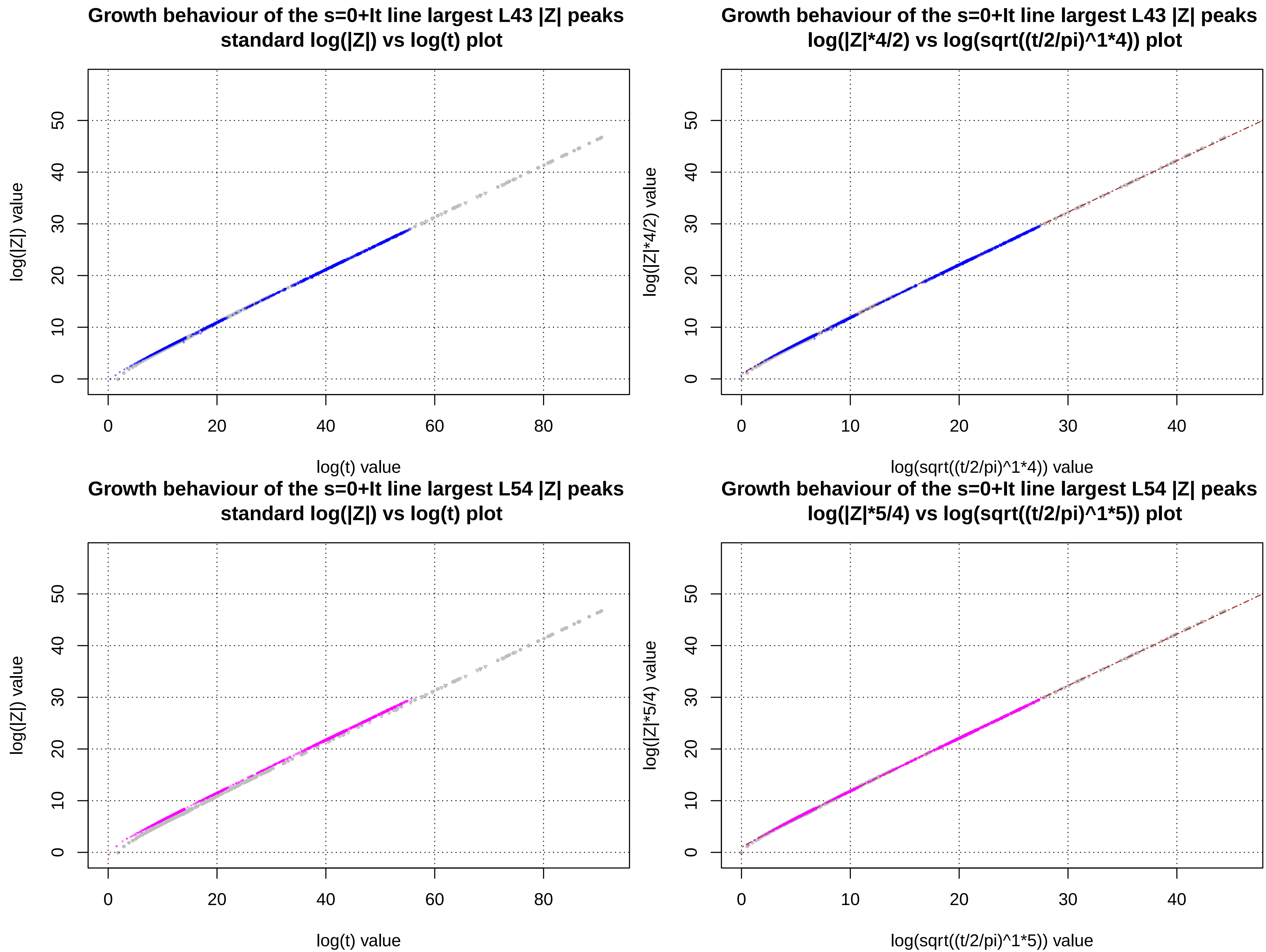

Figure 5: A graphical comparison for each investigated 1st degree L-function or Davenport-Heilbronn function of **left column** - $\log(|Z_L|)$ vs $\log(t)$, **right column** - $\log\left(|Z_L| \cdot \frac{N_C}{N_2.1656+log(seq(.1,50,by=.1)\chi\neq 0}\right)$ vs $\log\left(\sqrt{(\frac{t}{2\pi})^d \cdot N_C}\right)$ for top row - $L(\chi_4(3,.), 0.0 + I \cdot t)$ and bottom row - $L(\chi_5(4,.), 0.0 + I \cdot t)$. With one conservative fitted model for extreme peak height trend growth rate.

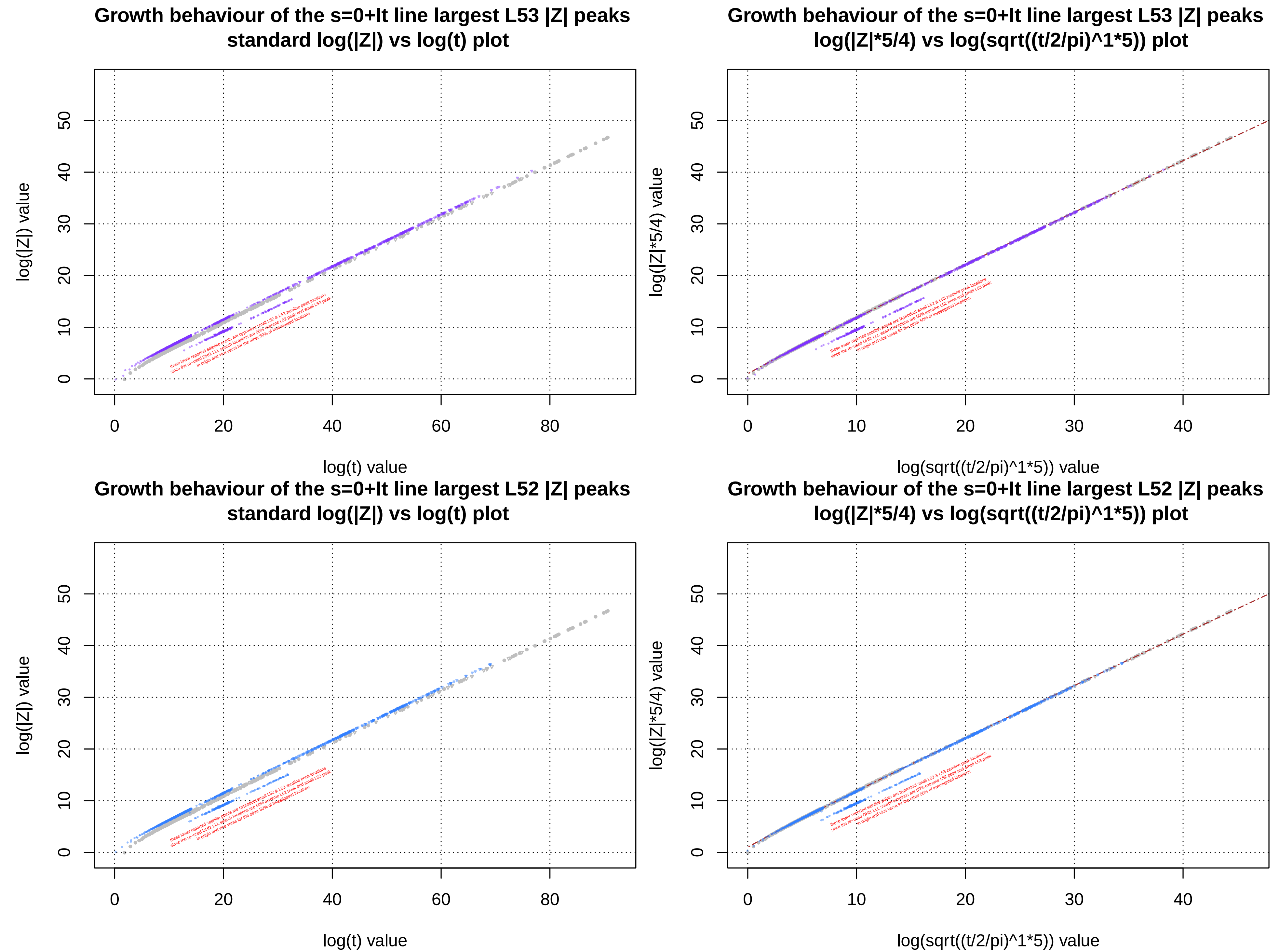

Figure 6: A graphical comparison for each investigated 1st degree L-function or Davenport-Heilbronn function of **left column** - $\log(|Z_L|)$ vs $\log(t)$, **right column** - $\log\left(|Z_L| \cdot \frac{N_C}{N_{\chi \neq 0}}\right)$ vs $\log\left(\sqrt{(\frac{t}{2\pi})^d \cdot N_C}\right)$ for top row - $L(\chi_5(3,.), 0.0 + I \cdot t)$ and bottom row - $L(\chi_5(2,.), 0.0 + I \cdot t)$. With one conservative fitted model for extreme peak height trend growth rate.

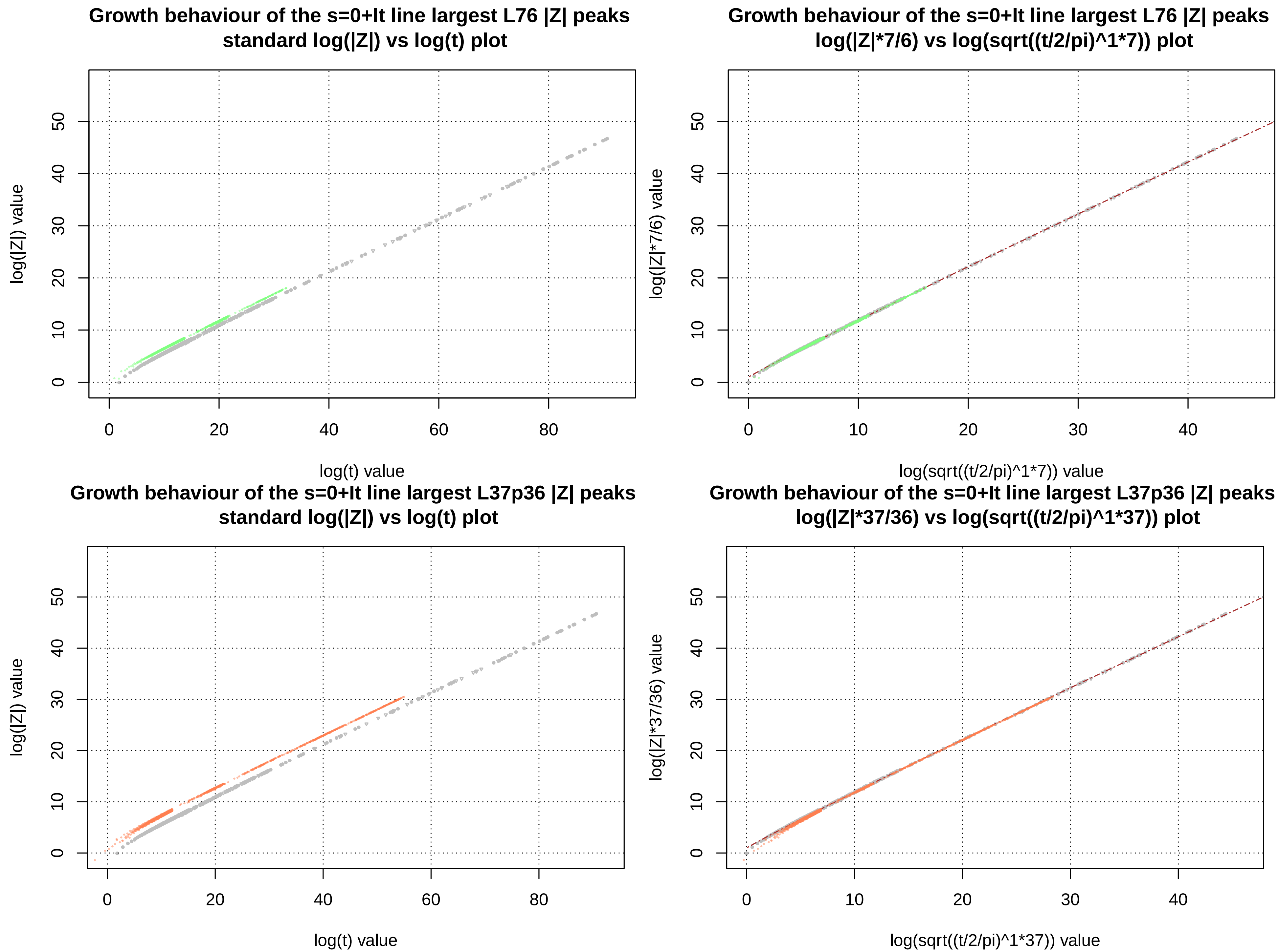

Figure 7: A graphical comparison for each investigated 1st degree L-function or Davenport-Heilbronn function of **left column** - $\log(|Z_L|)$ vs $\log(t)$, **right column** - $\log\left(|Z_L| \cdot \frac{N_C}{N_{\chi \neq 0}}\right)$ vs $\log\left(\sqrt{(\frac{t}{2\pi})^d \cdot N_C}\right)$ for top row - $L(\chi_7(6,.), 0.0 + I \cdot t)$ and bottom row - $L(\chi_{37}(36,.), 0.0 + I \cdot t)$. With one conservative fitted model for extreme peak height trend growth rate.

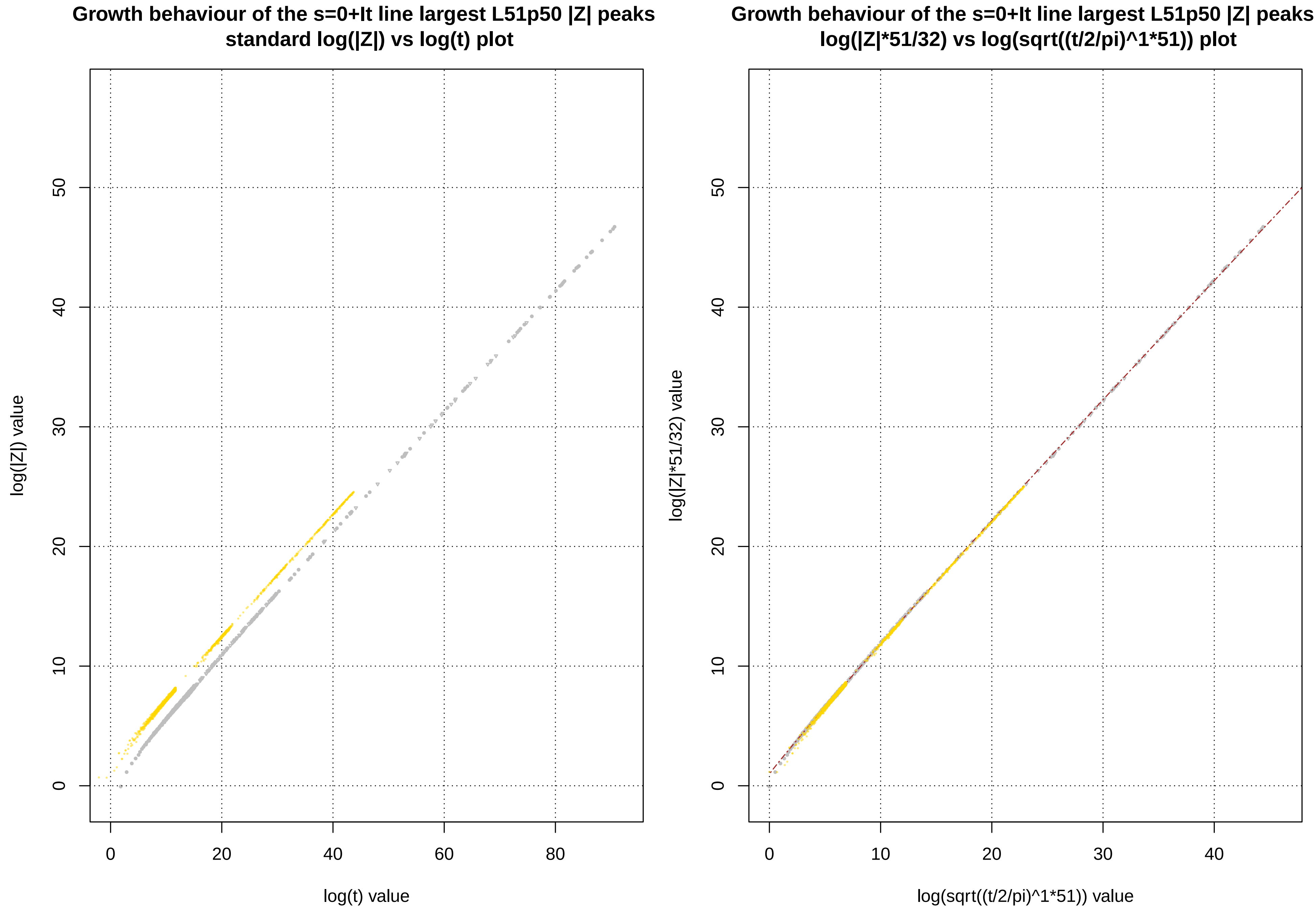

Figure 8: A graphical comparison for each investigated 1st degree L-function or Davenport-Heilbronn function of **left column** - $\log(|Z_L|)$ vs $\log(t)$, **right column** - $\log\left(|Z_L| \cdot \frac{N_C}{N_{\chi \neq 0}}\right)$ vs $\log\left(\sqrt{(\frac{t}{2\pi})^d \cdot N_C}\right)$ for $L(\chi_{51}(50,.), 0.0 + I \cdot t)$. With one conservative fitted model for extreme peak height trend growth rate.

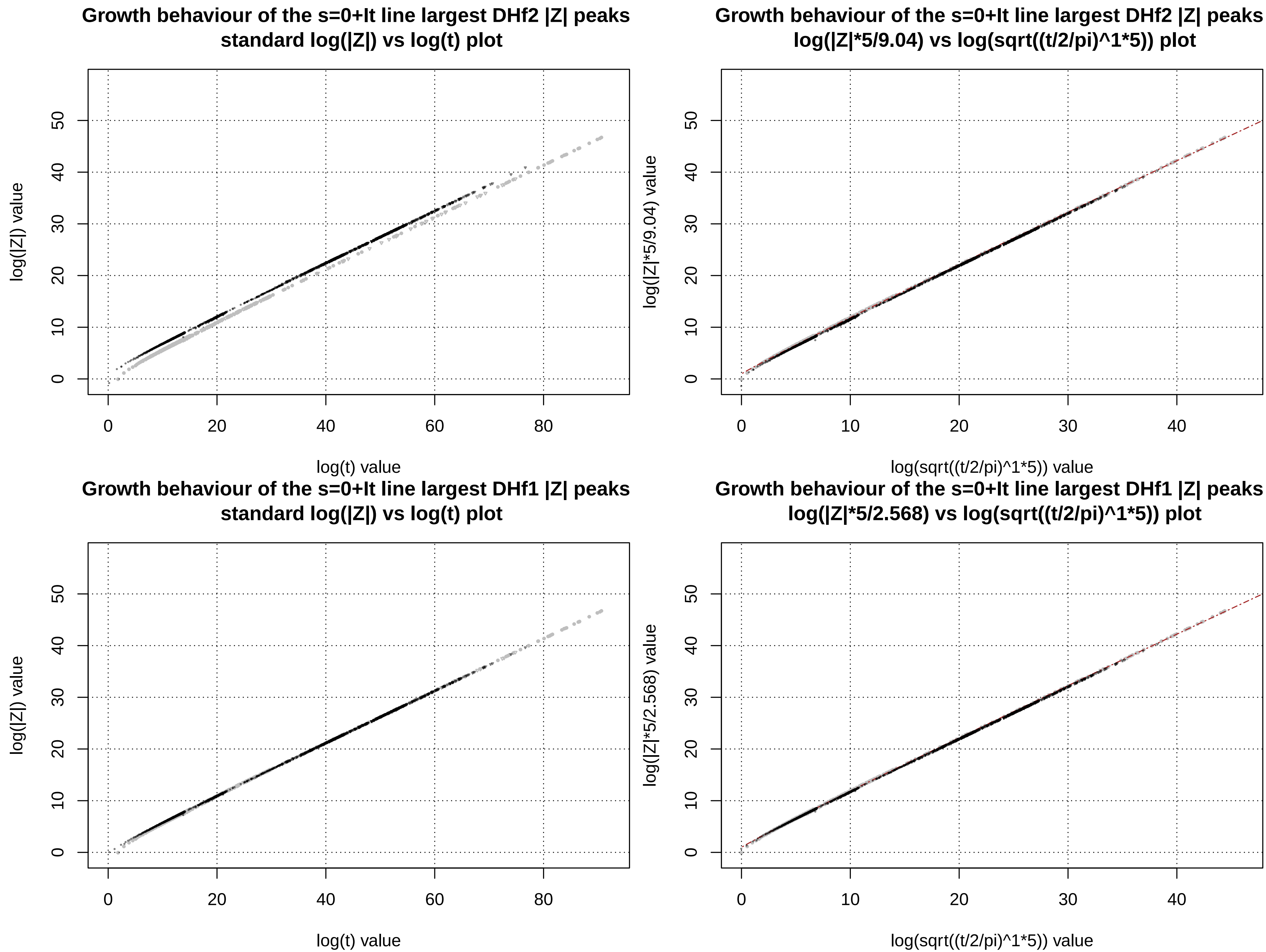

Figure 9: A graphical comparison for each 1st degree investigated 1st degree L-function or Davenport-Heilbronn function of **left column** - $\log(|Z_L|)$ vs $\log(t)$, **right column** - $\log\left(|Z_L| \cdot \frac{N_C}{N_{\chi \neq 0}}\right)$ vs $\log\left(\sqrt{(\frac{t}{2\pi})^d \cdot N_C}\right)$ for top row - $f_2(0.0 + I \cdot t)$ and bottom row - $f_1(0.0 + I \cdot t)$. With one conservative fitted model for extreme peak height trend growth rate.

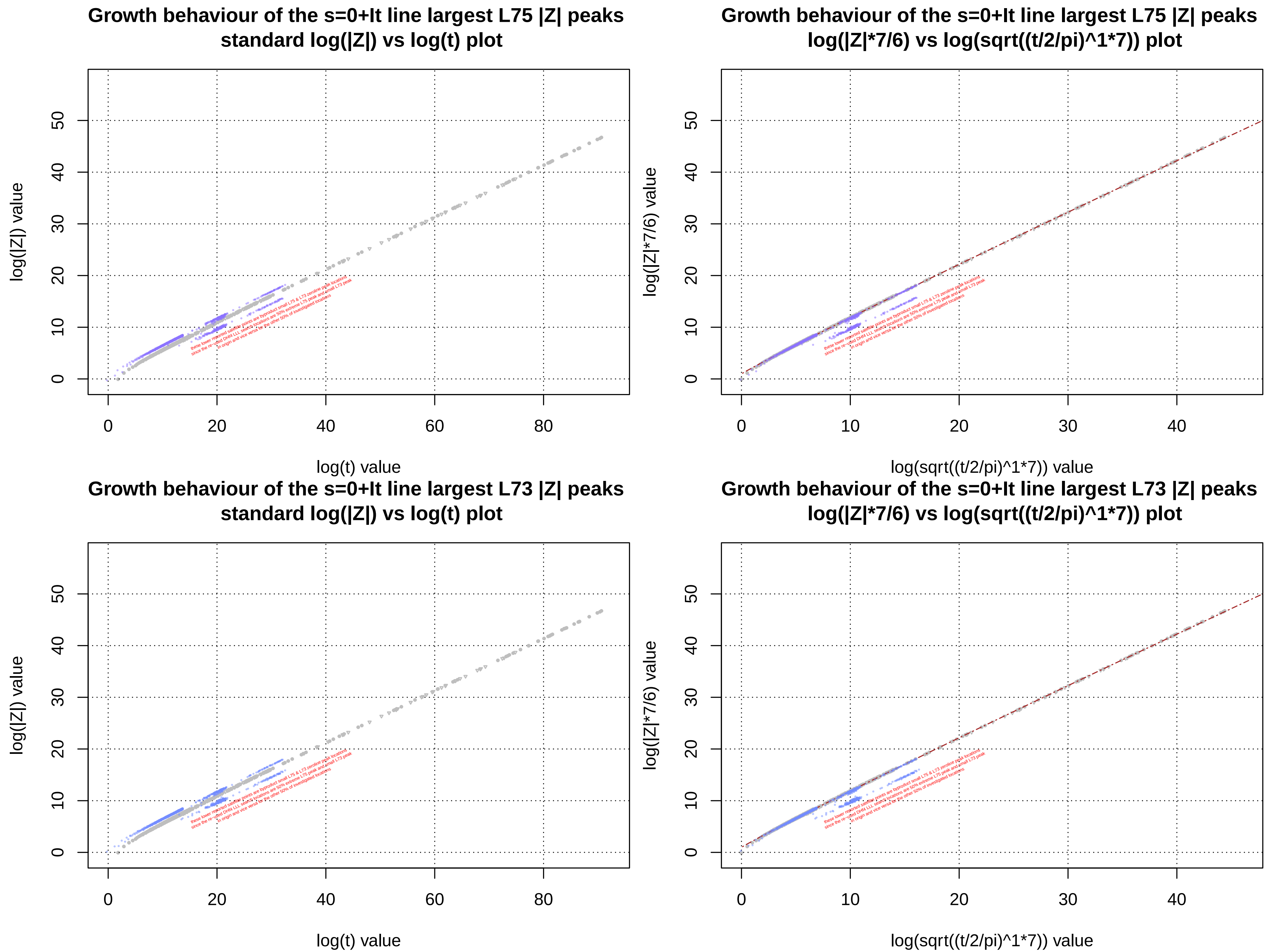

Figure 10: A graphical comparison for each investigated 1st degree L-function or Davenport-Heilbronn function of **left column** - $\log(|Z_L|)$ vs $\log(t)$, **right column** - $\log\left(|Z_L| \cdot \frac{N_C}{N_{\chi \neq 0}}\right)$ vs $\log\left(\sqrt{(\frac{t}{2\pi})^d \cdot N_C}\right)$ for top row - $L(\chi_7(5,.), 0.0 + I \cdot t)$ and bottom row - $L(\chi_7(3,.), 0.0 + I \cdot t)$. With one conservative fitted model for extreme peak height trend growth rate.

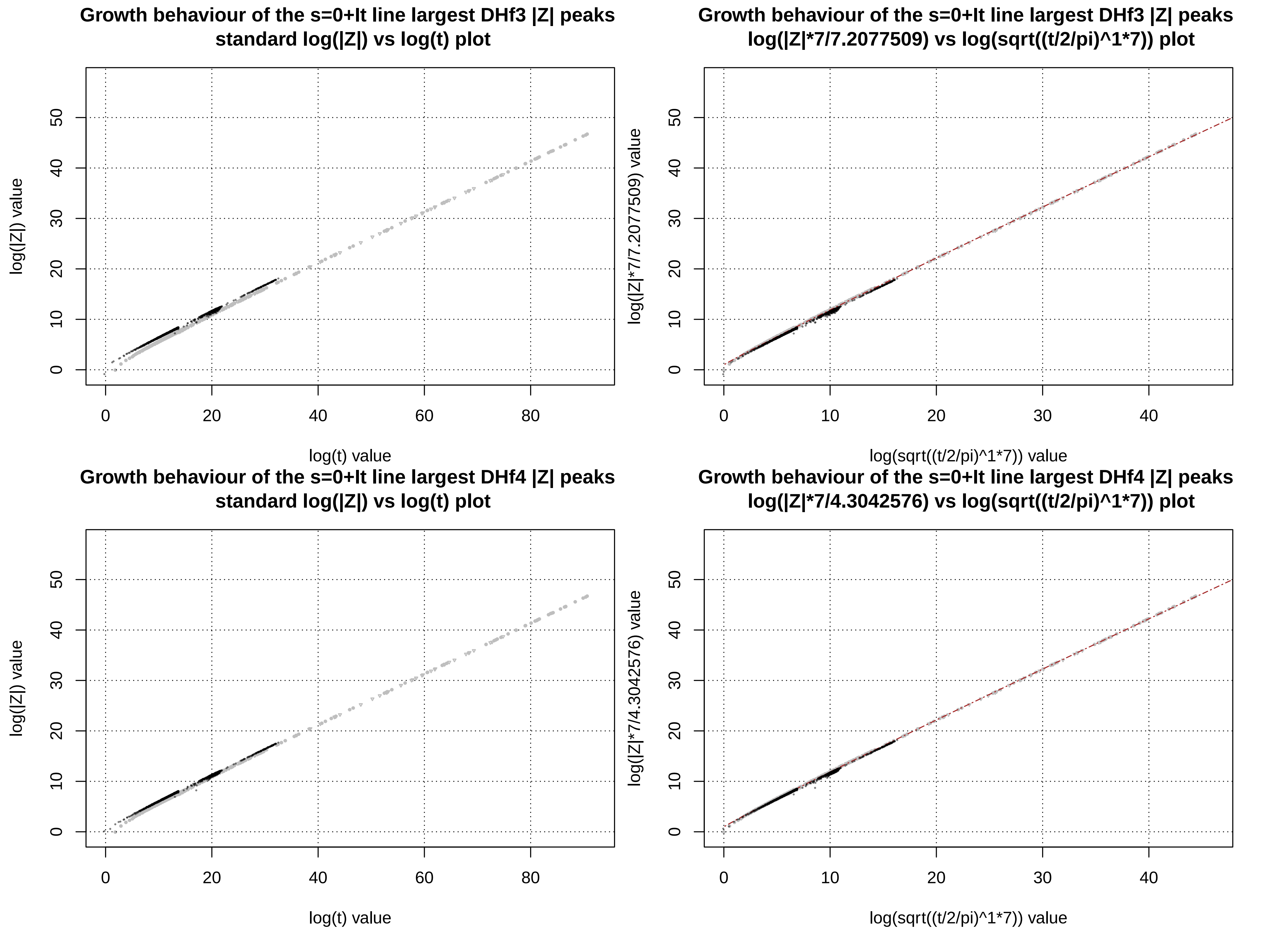

Figure 11: A graphical comparison for each investigated 1st degree L-function or Davenport-Heilbronn function of **left column** - $\log\left(|Z_L|\right)$ vs $\log(t)$, **right column** - $\log\left(|Z_L| \cdot \frac{N_C}{N_{\chi \neq 0}}\right)$ vs $\log\left(\sqrt{(\frac{t}{2\pi})^d \cdot N_C}\right)$ for top row - $f_3(0.0 + I \cdot t)$, and bottom row - $f_4(0.0 + I \cdot t)$. With one conservative fitted model for extreme peak height trend growth rate.

Highlights part 1

For the L- functions with only dirichlet characters of unity $|\chi| = 1$ or zero $|\chi| = 0$: figures 4, 5, 6, 7, 8 and 10:

In the left column, there is divergence between the L-function and Riemann Zeta |Z| extreme peak growth behaviour.

In the right column, except for the lowest t values the re-scaled L-function extreme growth behaviour to first order, tends to lie reasonably adjacent to the Riemann Zeta |Z| extreme peak growth behaviour. While the LLL search results t > 1e5 do find extreme peaks it doesn't collect all of them as the grid search does for lower t so there is still room for re-scaled extreme L-function peaks to get closer to the Riemann Zeta |Z| behaviour.

The re-scaling of the y-axis is a simple argument using the two bits of information (i) that the Riemann Zeta has only non-zero characters of unity ($\chi = 1$) while the other presented 1st degree L-functions have ($|\chi| = 0$ OR 1) and (ii) for extreme peak locations at high t, the **imaginary component** of each lowest lying dirichlet series term contributes approximately 1 to the dirichlet series sum.

The re-scaling of the x-axis is a recognition that each L-function has an intrinsic length $N_1 = \sqrt{\left(\frac{t}{2\pi}\right)^d \cdot N_C}$ and in properly comparing the growth behaviour of two L-functions the difference in their N_1 values must be taken into account. (The intrinsic length $N_2 = 2 \cdot \left(\frac{t}{2\pi}\right)^d \cdot N_C$ also exists and tapered Dirichlet series sums using this length can approach high precision L-function calculation performance for some L-functions.)

For the Davenport-Heilbronn functions in figures 9 and 11:

In the left column, there is divergence between the Davenport-Heilbronn and Riemann Zeta |Z| extreme peak growth behaviour.

In the right columns (comparing to the pure L-function figures), except for the lowest t values there is weak but possible evidence that the re-scaled Davenport-Heilbronn extreme growth behaviour **may be lower than** the Riemann Zeta |Z| extreme peak growth behaviour. In this case, the attempted y-axis re-scaling uses the $\sum_{k=1}^{N_C} |\chi_k|$ of the (absolute value of the) real coefficients of the known Davenport-Heilbronn dirichlet series.

Individual comparison of Ramanujan Tau L-function Riemann Siegel Z function extreme peak behaviour for $s = 5.5 + I \cdot t$ and $s = 6 + I \cdot t$ lines in the complex plane to the $|Z_\zeta(0 + I \cdot t)|$ and $|Z_\zeta(0.5 + I \cdot t)|$ extreme peak behaviour respectively.

To identify extreme Riemann Siegel |Z| function peak locations for the Ramanujan Tau L-function,

1. a grid search $\Delta = 0.05$ for $t < 65 \cdot (2 \cdot \pi)/N_C$ within each $t = [N \cdot (2 \cdot \pi)/N_C, (N+1) \cdot (2 \cdot \pi)/N_C)$ interval using estimation method (i) is conducted with some manual cleanup for peaks on the boundary edges between piecewise intervals

2. a grid search $\Delta = 0.05$ for $65 \cdot (2 \cdot \pi)/N_C \leq t < 650 \cdot (2 \cdot \pi)/N_C$ within each $t = [N \cdot (2 \cdot \pi)/N_C, (N+1) \cdot (2 \cdot \pi)/N_C)$ interval using estimation method (iii) with a method (i) calculation for the final identified peak location. Method (ii) using 128 point tapering gave poorer results than method (iii) using method (i) as reference so method (ii) was dropped for this L-function. Some investigation suggested than greater tapering is required with method (ii) for the Ramanujan tau L-function due to the non-periodic nature of its dirichlet character magnitudes.

3. a grid search $\Delta = 0.05$ for $650 \cdot (2 \cdot \pi)/N_C \leq t < 10001 \cdot (2 \cdot \pi)/N_C$ within each $t = [N \cdot (2 \cdot \pi)/N_C, (N+1) \cdot (2 \cdot \pi)/N_C)$ interval using estimation method (iii).

4. using LLL algorithm predictions for extreme peak locations

(I) a short interval grid search $\Delta = 0.05$ over a $\delta t = \pm 0.25$ interval about the LLL predicted location using estimation method (iii) is conducted

(II) a longer 32768 point fourier grid search $\Delta = 0.05$ over a $\delta t = [-1638.35, 1638.4]$ interval about the LLL predicted location using estimation method (iv) (spectral filtered Euler Product N_1 truncation) often conducted with a method (iii) calculation for the final identified peak location

(III) for $t \gtrsim 10^6$ a longer 32768 point fourier grid search $\Delta = 0.05$ over a $\delta t = [-1638.35, 1638.4]$ interval about the LLL predicted location using preliminary estimation method (v) (spectral filtered Euler Product with heavy N_{1sub} truncation) where $\beta_{tau} = 100, 10, 1$ hyperparameters values where used depending on the t value. Some peak heights were recalculated for $\beta_{tau} = 10$ results using $\beta_{tau} = 100$ (and $\beta_{tau} = 1$ results using $\beta_{tau} = 10$) as the higher accuracy method.

Adjusting the LLL search algorithm [33] constraints for the Ramanujan Tau L-function

Attempts at using the LLL search algorithm for the Ramanujan Tau L-function employed a variety of modifications to $\frac{t_{peak} * log(k)}{2\pi}$ constraints mainly based on known arccos $\left(\frac{tau(k)}{k^{5.5}} \cdot \frac{1}{2}\right)$ behaviour arising from Ramanujan's third conjecture $|\tau(p)| \leq 2p^{\frac{11}{2}}$ proved by Deligne [41]. However, the performance at identifying extreme peaks was clearly not as successful as for periodic 1st degree L-functions and hence drove the dramatic increase in the analysed fourier spectrum width (from $\delta t = [-6.35, 6.4]$ with periodic 1st degree L-functions to $\delta t = [-1638.35, 1638.4]$ with the Ramanujan Tau L-function) which helped locate nearby larger peaks.

Ramanujan Tau L-function results

Figure 12 displays a comparison of $|Z_\zeta|$ behaviour with located extreme Riemann Siegel Z function peaks for

top row - the lower boundary of the critical strip $s = 5.5 + I \cdot t$ and

bottom row - the critical line $s = 6 + I \cdot t$

versus the behaviour for the lower boundary of the Riemann Zeta function critical strip and its critical line respectively.

1. left column - $log\left(|L_\tau(s)|\right)$ vs $\log(t)$
2. right column - $log\left(|Z_L(s)|\tau \cdot \frac{N_C}{N_{\chi\neq 0}}\right)$ vs $\log\left(\frac{1}{d} \cdot \log\left(\sqrt{(\frac{t}{2\pi})^d}\right) + \log\left(\sqrt{N_C}\right)\right)$

where d=2, N_C=1 and $\frac{N_C}{N_{\chi\neq 0}} := \frac{1}{N_{\chi\neq 0}:=1} := 1$ is used for the Ramanujan Tau L-function.

For the top row,

- the Box Cox transformation based trend growth rate model given in equation (1) of this current paper is displayed shown, as brown dot-dashed line.
- a further transformation $\frac{2.1656}{2} + \frac{log\left(|Z_L(s)|\tau \cdot \frac{N_C}{N_{\chi\neq 0}}\right)}{2}$ is shown for some of the data points as green triangles.

For the bottom row,

- the Box Cox transformation based fitted trend growth rate model (t < 1e31) from [21] for the critical line of the Riemann Zeta function

$$log\left[|Z_\zeta(0.5 + I \cdot t)|_{max}\right]_{trend} \approx 1.274014 + 0.866363 \cdot \left(\frac{\sqrt{\log\left[\sqrt{\frac{t}{2\pi}}\right]} - 1}{0.5}\right) \tag{5}$$

is displayed, shown as brown dot-dashed line.

The bottom row panels in figure 12 contain a much wider set of extreme peaks (and lesser peaks) than was reported in [21].

Highlights part 2

For the Ramanujan Tau L-function

Examining the top row panels of figure 12,

- the growth rate of extreme peaks on the lower boundary of the Ramanujan Tau L-function critical strip clearly exceeds the growth rate of extreme peaks on the lower boundary of the Riemann Zeta function critical strip.
- the green triangle behaviour in the righthand panel of the top row estimates the growth rate of |Z| extreme peaks along $s = 5.5 + I \cdot t$ for the Ramanujan Tau L-function as roughly the square of the $|Z_{zeta}(s = 0 + I \cdot t)|$ extreme peak growth rate.

Examining the bottom row panels of figure 12,

- the wider range of extreme peaks investigated in this paper for the Ramanujan Tau L-function on its critical line continues to exhibit to first order the same extreme peak growth rate as $|Z_{zeta}(s = 0.5 + I \cdot t)|$

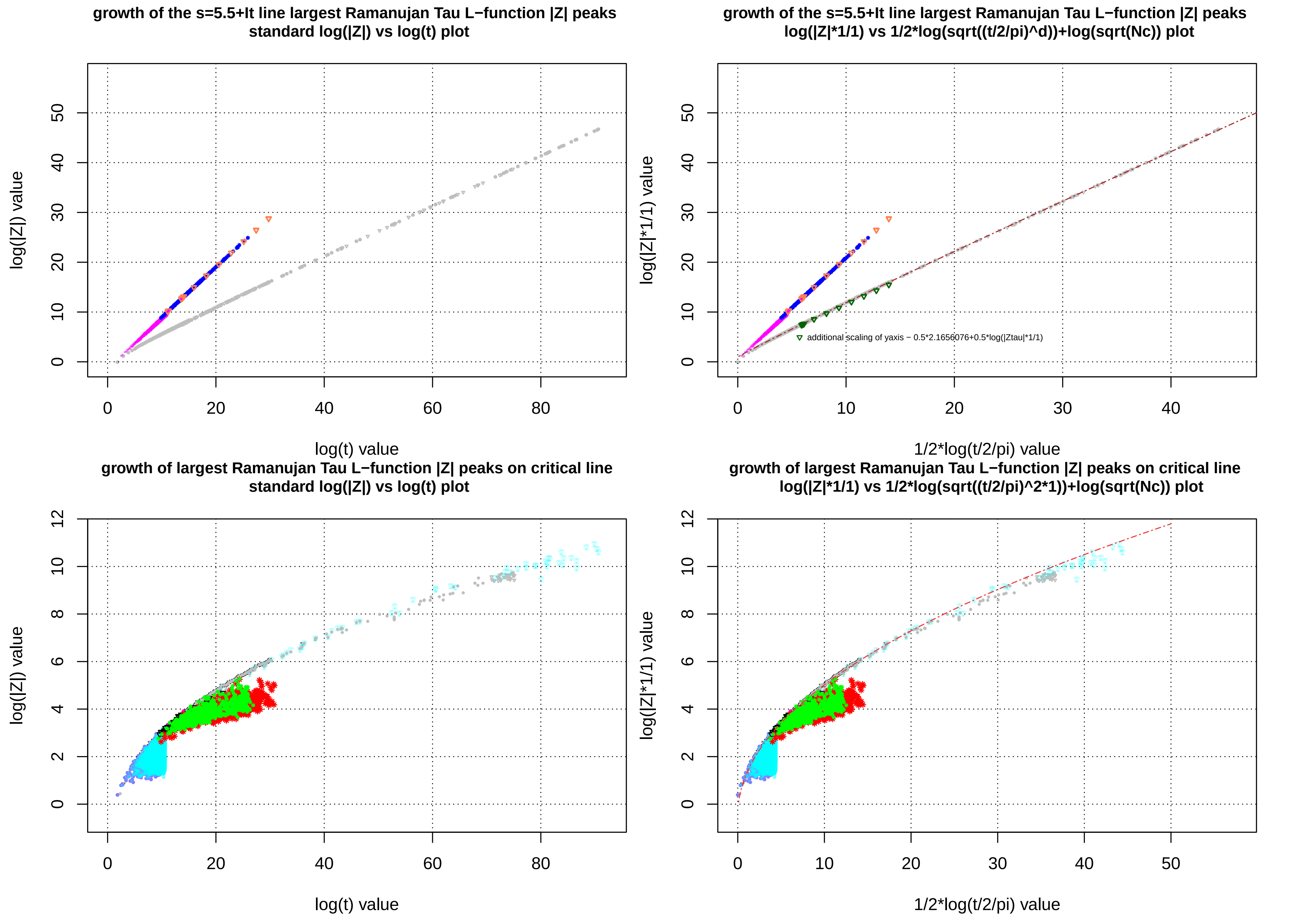

Figure 12: A graphical comparison for the 2nd degree Ramanujan's Tau L-function and the Riemann Zeta function extreme peaks of **left panel** - $\log\left(|Z_L|\right)$ vs $\log(t)$, **right panel** - $\log\left(|Z_L| \cdot \frac{N_C}{N_{\chi\neq 0}}\right)$ vs $\frac{1}{d} \cdot \log\left(\sqrt{(\frac{t}{2\pi})^d}\right) + \log\left(\sqrt{N_C}\right)$ for **top row** - $s = 5.5 + I \cdot t$ (zero line), and **bottom row** - $s = 6 + I \cdot t$ (critical line). The x axis rescaling in the right panel is different in figure 12 compared to figures 4-11 because the two L-functions being compared in figure 12 are of different degrees. With one a different fitted model for extreme peak height trend growth rate depending on Re(s). Riemann Zeta function behaviour shown in gray for known peak heights (and cyan for high t preliminary peak height estimates.

Testing equation (1) and (3) trend model estimates by assessing their performance at predicting $s = 1 + I \cdot t$ behaviour

A second available method of assessing equation (1) and (3) trend model performance is by co-opting the Riemann Zeta functional equation to extract $s = 1 + I \cdot t$ extreme peak trend behaviour predictions and compare such predictions to observed $s = 1 + I \cdot t$ behaviour. Figure 13 displays this comparison as well as mapping the exact $s = 0 + I \cdot t$ peak heights to the $s = 1 + I \cdot t$ line shown as gray dots.

Given the Riemann Zeta functional equation [22-26]

$$\begin{aligned}
\zeta(s) &= \chi(s)\zeta(1-s) && (6)\\
&= 2^s\pi^{s-1} sin(\frac{\pi s}{2})\Gamma(1-s)\zeta(1-s) && (7)\\
&\equiv e^{\log\left[\chi(s)\right]}\zeta(1-s) && (8)\\
&= e^{\left((s-1/2)\cdot\log[Pi]+\log\Gamma\left[(\frac{(1-s)}{2})\right]-\log\left[\Gamma(\frac{s}{2})\right]\right)}\zeta(1-s) && (9)\\
&\equiv e^{-2\theta_{ext}(s)}\zeta(1-s) && (10)
\end{aligned}$$

where

$$\chi(s) = e^{\left((s-1/2)\cdot\log[Pi]+\log\Gamma\left[(\frac{(1-s)}{2})\right]-\log\left[\Gamma(\frac{s}{2})\right]\right)} \tag{11}$$

from equation (9) provides the most numerically stable version of calculating $\chi(z)$ for the Pari-GP [34] calculations performed in this paper.

Transformation of equations (1) and (3) to $s = 1 + I \cdot t$ predictions involves the mapping

$$\left[|Z_\zeta(1+I\cdot t)|_{max}\right]_{trend} \sim |\chi(1+I\cdot t)|\cdot\exp\left[2.1656076 + 1.1459223\cdot\left(\frac{\left(\log\left[\sqrt{\frac{t}{2\pi}}\right]\right)^{0.9604378} - 1}{0.9604378}\right)\right] \tag{12}$$

and

$$\left[|Z_\zeta(1+I\cdot t)|_{max}\right]_{\text{trend}} \sim |\chi(1+I\cdot)|\cdot\exp\left[2.1656 + \log\left(\sqrt{\frac{t}{2\pi}}\right)\right] \tag{13}$$

as shown when comparing equations (1) (brown dot-dashed line) and (3) (green dashed line) based estimates to predict $s = 1 + I \cdot t$ in figure 13.

Highlights part 3

1. The close agreement between the gray dots and the coloured data points shows that the $s = 0 + I \cdot t$ peak heights are accurately mapped to the $s = 1 + I \cdot t$ peak heights via the functional equation behaviour.
2. Equation (12) shows reasonable performance as a trend estimate for t < 1e18 but is inaccurate for t > 1e18

3. Equation (13) is an unphysical trend estimate predicting a constant extreme peak height for the $s = 1 + I \cdot t$ line

Numerically it is easy to see that

$$|\chi(1 + I \cdot t)| \to \frac{1}{\sqrt{\frac{t}{2\pi}}} \text{ for } t > \pi \tag{14}$$

with four decimal place accuracy and rapidly improving.

Hence re-examining equation (13)

$$\begin{aligned} \left[|Z_\zeta(1 + I \cdot t)|_{max}\right]_{\text{trend}} &\sim |\chi(1 + I \cdot t)| \cdot \exp\left[2.1656 + \log\left(\sqrt{\frac{t}{2\pi}}\right)\right] \\ &\to \frac{1}{\sqrt{\frac{t}{2\pi}}} \cdot \exp[2.1656] \cdot \sqrt{\frac{t}{2\pi}} \qquad (15) \\ &= \exp[2.1656] \qquad (16) \end{aligned}$$

shows that constant amplitude behaviour of extreme peaks on $s = 1 + I \cdot t$ is the predicted (unphysical) behaviour of equation (3) when mapped to the $s = 1 + I \cdot t$ line.

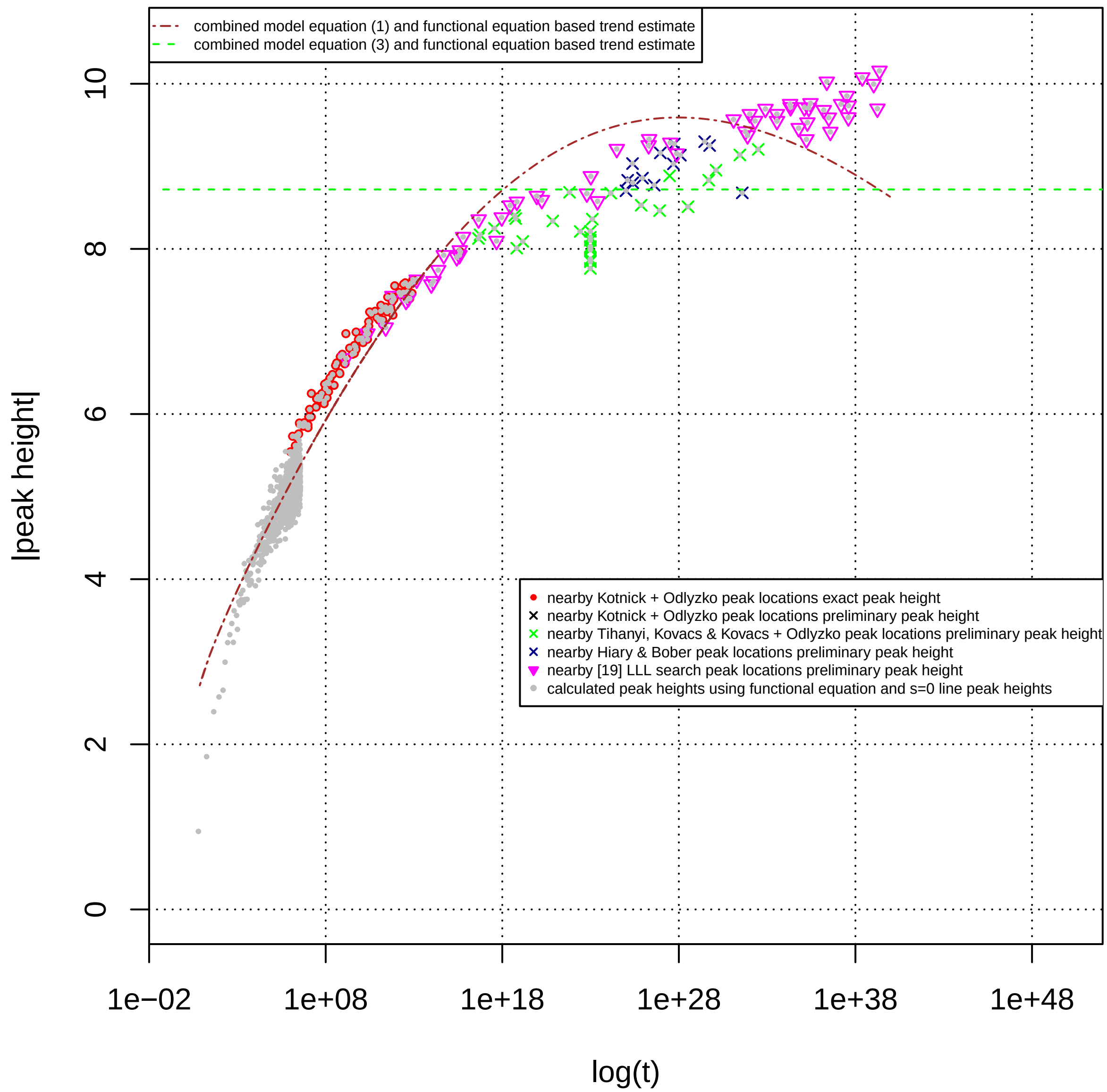

Figure 13. Observed |Z| versus log(t) scatterplot of (i) good precision extreme Riemann Siegel Z function peak heights on the $s = 1 + I \cdot t$ line between 1e6 < t < 1e13 (red solid circles) equation (16) and (ii) preliminary extreme Riemann Siegel Z function peak heights between 1e6 < t < 1e40 based on spectral filtered heavily truncated Euler products based approximation method (v) overlayed by (I) dot-dashed brown line and green dashed line of two mapped trend estimates based on fitted $s = 0 + I \cdot t$ extreme peak behaviour (equations (1) and (3)) and the known functional equation and (II) shonw as gray points $s = 0 + I \cdot t$ extreme peak heights transformed by the known functional equation

Conclusions

A method for re-scaling the growth behaviour of extreme Riemann Siegel Z function of 1st degree L-functions with non-zero dirichlet characters of magnitude unity $|\chi| = 1$ is demonstrated to work on the $s = 0 + I \cdot t$ in addition to previously reported critical line behaviour. Using a mapping from the $s = 0 + I \cdot t$ line to the $s = 1 + I \cdot t$ line, two simple candidate trend growth formula for extreme Riemann Siegel Z function peaks of the Riemann Zeta function fitted to $s = 0 + I \cdot t$ data are shown to have obvious deficiencies.

Supplementary Material

A zip file of the located (extreme or otherwise) peak locations for the investigated L- functions and Davenport_Heilbronn counterparts is available on github https://github.com/johnpdmartin/sampling-investigations/blob/master/files_with_extreme_peaks_of_L_functions_on_zeroline.zip.

The typical format is 4-7 columns for files containing calculations by this author:

column 1 - t value (accurate to 0.05),

column 2 - N_1 value,

column 3 - residual of N_1 value (used for investigating multi-modal distribution of peaks within [N,N+1) piecewise intervals.

column 4 - |Z| value,

column 5 - Z value (sometimes calculated by more accurate method as crosscheck)

column 6 - text message describing method used for column 4 and 5

column 7 - may contain a number indicating element number of vector of estimates where peak occurred.

For files containing data from references there are typically column headers.

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